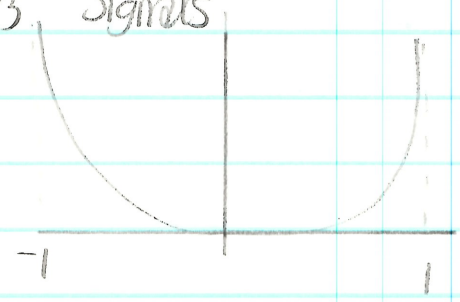
07/07/23 Signals

Example



$$\langle x(t), y(t) \rangle = \int_{-1}^{1} w(t)\, x(t)\, y(t)\, dt$$

Let x and y be random variables with the joint pdf $f_{xy}(x,y)$.

We know $f_{xy}(x,y) \geq 0$

We then define an inner product

$$\langle x, y \rangle = \iint_{-\infty}^{\infty} xy\, f_{xy}(x,y)\, dx\, dy = E[xy]$$

Then we can say things like $E[xy] = 0$, then x and y are orthogonal.

I Banach Spaces and Hilbert Spaces

Def: A complete normed vector space is a Banach space

Def: A complete normed vector space where the norm is induced from an inner product is a Hilbert space.

Complete: Convergence of an infinite sequence

A) Example: x_n: series of finite numbers

$x_1 = 2$

$x_2 = 2.4$ The series is defined s.t.

$x_3 = 2.41$

$x_4 = 2.414$ $\lim_{n\to\infty} x_n = \sqrt{2}$

$\vdots$

Now lets look at

$x_n - x_m$: $x_4 - x_3 = 0.004$

$\downarrow$

As m and n get bigger, the difference gets smaller

If $\lim_{\substack{n\to\infty \\ m\to\infty}} |x_n - x_m| = 0$ we can then say $\{x_n\}$

converges according to the Cauchy criterion.

c) Examples

$(C[a,b], d_\infty)$ is a Banach space

$(C[a,b], L_p)$ is not a Banach space

$(L_p[a,b], L_p)$ is a Banach space

$(L_2[a,b], L_2)$ is a Hilbert space

II Orthogonal Subspaces

Def: Let V be an inner product space.
Let W_1 be a subspace of V, and let W_2 be a subspace of V. W_1 and W_2 are orthogonal if $\forall x_1 \in W_1$ and $x_2 \in W_2$

$$\langle x_1, x_2 \rangle = 0$$

Def: Let V be a vector space. A subset $W_1 \subset V$ is a subspace if W_1 is itself a vector space.

Def: For a subspace $W_1 \subset V$, the set of all vectors in V orthogonal to W_1 is called the orthogonal compliment. It is denoted as $W_1^\perp$.

Fact: The orthogonal complement of a subspace is a subspace.

Proof: Let $W_1 \subset V$ be a subspace
Let $W_2 = W_1^\perp$

$0 \in W_1^\perp$ by definition.

To show closure under linear combination.

Let $x_1, x_2 \in W_2$

$a_1 x_1 + a_2 x_2$ in W_2? Yes, for any $x \in W_1$

$$\langle a_1 x_1 + a_2 x_2, x \rangle = a_1 \langle x_1, x \rangle + a_2 \langle x_2, x \rangle$$
$$= 0 + 0$$

Theorem: Let W_1 and W_2 be subspaces of vector space V.

- $W_1^\perp$ is a closed subspace of V
- $W_1 \subset (W_1^\perp)^\perp$
- If $W_2 \subset W_1$ then $W_1^\perp \subset W_2^\perp$
- If $x \in W_1 \cap W_1^\perp$, then $x = 0$
- $V^\perp = \{0\}$ and $0^\perp = \{V\}$

III Linear Transformation

Def Let X and Y be vector spaces.

A transformation $L: X \to Y$ is a linear transformation if $\forall\, x_1, x_2, x_3 \in X$:

- $L(\alpha x_1) = \alpha L(x_1)$
- $L(x_1 + x_2) = L(x_1) + L(x_2)$

A) Example

$X = C[0, \infty) \quad Y = X$

$x(t) \in X$

$$Lx(t) = \int_0^t h(\tau)\, x(t-\tau)\, d\tau$$

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